

A REPORT OF SIX WEEK TRAINING (TR-102)

at

ICE Technology Lab.

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF

DEGREE OF

BACHELOR OF TECHNOLOGY

(Computer science and Engineering)



JUNE-JULY , 2025

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CERTIFICATE BY ICE Technology Lab.

This is to certify that **Sidak Arora** has successfully completed Six week training at ICE Technology Lab in “**web development**” during the period from **28th June 2025** to **25th July 2025**, in partial fulfillment of the requirements for the award of the degree of **B.Tech. (Computer Science and Engineering)** at **Guru Nanak Dev Engineering College, Ludhiana**. The training report submitted to the Department of Computer Science and Engineering is an authentic record of the work conducted during the training.

Signature & Seal: Date: _____ Date : _____

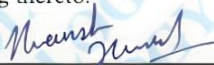


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This is to certify that
Mr./Mrs. SIDAK ARORA
was a full time Student of **4 WEEKS** Certification has satisfactorily
Completed the industrial training and having secured a grade of A is therefore awarded this
INDUSTRIAL TRAINING
in
WEB DESIGNING(CSS JAVASCRIPT HTML)
with all the rights, honors and privileges of having pertaining thereto.


Director
Technology Expert Committee


President
Course & Evaluation Committee
Declared on : **28.07.2025**

Score	Grade	Significance
9.00-10.00	A+	Outstanding
8.00-8.99	A	Excellent
7.00-7.99	B+	Very Good
6.00-6.99	B	Good
5.00-5.99	C	Satisfactory
4.00-4.99	D	Poor


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CANDIDATE'S DECLARATION

I, **Sidak Arora**, hereby declare that I have undertaken four week training “**Web Development**” during the period from **28th June 2025** to **25th July 2025** in partial fulfillment of requirements for the award of the degree of **B.Tech. (Computer Science and Engineering)** at **Guru Nanak Dev Engineering College, Ludhiana**. The work presented in this training report, submitted to the Department of Computer Science and Engineering at Guru Nanak Dev Engineering College, Ludhiana, is an authentic record of the training work.

The Six week industrial training Viva–Voce Examination of _____ has been held on _____ and accepted.

Signature of Internal Examiner

Signature of External Examiner

ABSTRACT

This report presents the details of my one-month industrial training undertaken at **ICE Technology Lab, Patna** from **28th June to 25th July, 2025**, where I worked on a real-time project titled “**Portfolio**” The objective of this project was to design and develop a smart, user-friendly, full-stack web application that enables students and professionals to create professional, ATS-compliant resumes with the help of Artificial Intelligence.

The application was built using **HTML, JavaScript, CSS**, for the front-end

During the training, I followed an agile approach with weekly goals, learning and applying various tools and concepts including authentication, responsive design, PDF export, and AI prompt engineering. This project helped me enhance my technical skills, practical knowledge of industry tools, and understanding of building scalable web applications. The experience has laid a strong foundation for my future aspirations in web development and AI-based solutions.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **ICE Technology Lab, Patna**, for providing me the opportunity to undergo one-month industrial training from **28th June to 25th July, 2025**. The training experience was immensely valuable and gave me practical exposure to full-stack development and AI integration in a real-world setting.

I am especially thankful to my mentors and trainers at ICE Technology Lab for their constant guidance, encouragement, and technical support throughout the development of my project, **“Portfolio”**. Their expertise, timely feedback, and approachable attitude made the learning process engaging and fruitful.

I also extend my heartfelt thanks to the **faculty and training coordinators of Guru Nanak Dev Engineering College, Ludhiana**, for encouraging us to explore real-time industry projects as part of our academic curriculum. Their continued support and mentorship were crucial in shaping my understanding of practical software development.

I am grateful to my peers and fellow trainees who made the training experience collaborative and enriching through active knowledge sharing and teamwork. Lastly, I would like to thank my family and friends for their unwavering encouragement and moral support, which motivated me throughout this journey.

This training has been a stepping stone in my professional development, equipping me with the skills and confidence to pursue my goals in the field of web development and artificial intelligence.

About ICE Technology Lab

ICE Technology Lab is a leading IT training and software development company based in Mithulok, West Boring Canal Road, Patna, Bihar. The organization is committed to delivering high-quality, industry-relevant training programs in emerging technologies such as web development, artificial intelligence, cloud computing, data science, cybersecurity, and mobile app development. With a vision to bridge the gap between academic knowledge and industrial requirements, ICE Tech Lab plays a vital role in shaping the careers of aspiring developers, engineers, and IT professionals.

The company offers hands-on project-based training under the mentorship of industry experts, ensuring that learners gain both theoretical clarity and practical experience. Students work on real-time, client-oriented projects, which enhances their understanding of software design, architecture, and deployment processes. The training methodology is designed around agile principles, collaborative learning, and outcome-based education to prepare trainees for real-world challenges.

ICE Technology Lab maintains a strong focus on modern tech stacks, including frameworks like Next., FRONTEND, Node., along with integrations for cloud platforms, AI APIs, and secure payment systems. Trainees are exposed to tools such as Firebase, Cloudinary, Razorpay, and AI frameworks like Genkit paired with Google's Gemini models, promoting innovation and technical proficiency.

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CHAPTER 1

INTRODUCTION

1.1 Basic Introduction:

Front-end development focuses on the part of a website or web application that users directly interact with. It involves creating the layout, design, and interactive features that make a website functional and visually appealing. Front-end developers use technologies such as HTML, CSS, and JavaScript to build user interfaces that are both responsive and user-friendly.

During my training, I learned and practiced FRONTEND, one of the most popular JavaScript libraries for building modern web applications. FRONTEND allows developers to create reusable components, manage data efficiently using state and props, and build dynamic, single-page applications.

Through this training, I gained a practical understanding of how front-end technologies work together — how HTML structures content, CSS styles the layout, JavaScript adds interactivity, and FRONTEND enhances performance and scalability. This knowledge helped me understand the core principles behind developing modern, responsive web interfaces.

Key Learning

1. HTML: Learned to structure web pages using semantic elements and forms.
2. CSS: Gained knowledge in designing responsive layouts using Flexbox and Grid.
3. JavaScript: Understood core concepts such as DOM manipulation, ES6 features, and event handling.

4. FRONTEND: Built interactive UIs, learned about components, props, states, hooks, and routing.
5. GitHub: Learned how to use Git commands for version control and collaborate on projects.

Background

- **HTML (HyperText Markup Language)** : It provided a simple way to structure text and add links to other documents, forming the very backbone of a webpage. Early websites were purely informational, like digital brochures with basic text and links.
- **CSS (Cascading Style Sheets)** : As the web grew, so did the desire to make it look better. This was a game-changer, allowing for more creative layouts, colors, and fonts. The first version of CSS was officially released in **1996**.
- **JavaScript** : This scripting language brought interactivity to websites, enabling things like pop-up alerts, form validations, and dynamic content changes without needing to reload the entire page.

1.2 Objectives:

The main objective of using FRONTEND for frontend development is to build fast, scalable, and maintainable user interfaces by breaking them down into small, reusable pieces called **components**. It simplifies the process of creating complex, interactive applications that can efficiently update and render data without reloading the entire page.

Build Reusable and Modular UIs

The primary goal of FRONTEND is to allow developers to create user interfaces using a component-based architecture. Think of components as custom, reusable HTML elements or LEGO bricks. You can build a small, isolated component like a search bar or a button and then reuse it throughout your application.

This approach makes your code:

- Maintainable: Updating a component in one place updates it everywhere it's used.
- Organized: Complex UIs are broken down into smaller, manageable files.
- Scalable: It's easy to add new features by creating new components.

Create Fast and Performant Applications

FRONTEND aims to maximize performance by using a Virtual DOM. Instead of directly manipulating the slow, browser-based DOM (Document Object Model), FRONTEND keeps a lightweight copy of it in memory.

When the application's data changes, FRONTEND:

1. Creates a new virtual DOM tree.
2. Compares it with the previous one to find the differences (a process called "diffing").
3. Calculates the most efficient way to update the real DOM and applies *only* the necessary changes.

This process minimizes slow DOM operations, resulting in a much faster and smoother user experience, especially in applications with a lot of data updates.

Simplify Complex State Management

For dynamic applications, managing how data changes and flows through the UI (the "state") can become very complex. FRONTEND provides and integrates seamlessly with powerful state management libraries (like Redux or Zustand).

The objective is to create a predictable state flow, making it easier to understand how data affects the UI, debug issues, and build large-scale Single-Page Applications (SPAs) where the user interface updates dynamically without full page reloads.

1.3 Scope:

The future for those trained in FRONTEND looks incredibly bright and secure. As businesses increasingly prioritize dynamic, high-performing, and scalable web and mobile applications, the demand for skilled FRONTEND developers is set to remain strong. Here's a breakdown of the future scope of FRONTEND training.

High Demand and Abundant Career Opportunities

FRONTEND continues to be one of the most popular and in-demand JavaScript libraries in the web development landscape. This is due to several key factors:

- **Wide Adoption:** From tech giants like Meta, Netflix, and Airbnb to countless startups, FRONTEND is a go-to choice for building modern user interfaces. This widespread adoption translates to a consistent and growing demand for developers with FRONTEND expertise.
- **Versatile Career Paths:** FRONTEND training opens doors to a variety of roles, including:
- **Frontend Developer:** Specializing in creating engaging and interactive user interfaces.
- **Full-Stack Developer:** Combining FRONTEND with backend technologies like Node. to build complete web applications.
- **FRONTEND Native Developer:** Using FRONTEND principles to build native mobile apps for both iOS and Android.
- **UI/UX Engineer:** Focusing on the user experience and visual design implementation.

- **Competitive Salaries:** The high demand for FRONTEND skills translates into competitive salaries, even for entry-level positions, with significant growth potential as you gain experience.

Evolving with Future Web Development Trends

FRONTEND is not stagnant; it's continuously evolving to meet the demands of modern web development. Your training will position you to be at the forefront of these exciting trends:

- **Server Components:** A major shift in FRONTEND that allows for better performance by rendering components on the server. This reduces the amount of JavaScript sent to the client, leading to faster load times and improved user experience.
- **AI and Machine Learning Integration:** The future of web development will see a deeper integration of AI and machine learning. FRONTEND's component-based architecture makes it well-suited for building intelligent and personalized user experiences.
- **Micro-Frontends:** As applications become more complex, the ability to break them down into smaller, independently deployable frontends is crucial. FRONTEND's modular nature is perfect for this architectural style.
- **Progressive Web Apps (PWAs):** PWAs offer a native app-like experience on the web. FRONTEND is an excellent choice for building fast, reliable, and engaging PWAs.
- **WebAssembly Integration:** For performance-critical applications, WebAssembly allows for running code at near-native speed in the browser. FRONTEND can seamlessly integrate with WebAssembly modules.

1.4 Relevance to Academic Curriculum :

The FRONTEND. training is highly relevant to the academic curriculum, especially for students pursuing **Computer Science, Information Technology, or Software Engineering**. It complements theoretical knowledge with practical skills in **front-end web development**, which is often part of the syllabus under subjects like Web Technologies, Internet Programming, or Software Development.

Key points of relevance:

1. **Bridges Theory and Practice:** Concepts like HTML, CSS, and JavaScript taught in class are applied in real-world projects using FRONTEND..
2. **Modern Web Development Skills:** FRONTEND. is widely used in the industry, so learning it aligns with current trends in software development.
3. **Project-Based Learning:** Students gain experience in building interactive and responsive web applications, which enhances their project work and lab assignments.

CHAPTER 2

TRAINING WORK UNDERTAKEN

2.1 INTRODUCTION TO HTML

What is HTML?

HTML (HyperText Markup Language) was created by Tim Berners-Lee in 1991 as a standard for creating web pages. It's a markup language used to structure content on the web, defining elements like headings, paragraphs, links, and images. HTML forms the backbone of web content. In layman's terms, HTML is like the skeleton of a website. It's a set of instructions that tells a web browser how to display text, images, videos, and other elements on a webpage. Think of it as the building blocks that create the structure and look of a website, similar to how bricks and mortar are used to build a house.

- HTML is the language of the web, used to create websites.
- HTML defines the barebone structure or layout of web pages that we see on the Internet.
- HTML consists of a set of tags contained within an HTML document, and the associated files typically have either a .html or .htm extension.
- There are several versions of HTML, with HTML5 being the most recent version.

Features of HTML

It is platform-independent. For example, Chrome displays the same pages identically across different operating systems such as Mac, Linux, and Windows.

- Images, videos, and audio can be added to a web page (For example - YouTube shows videos on their website).
- HTML is a markup language and not a programming language.
- It can be integrated with other languages like CSS, JavaScript, etc. to show interactive (or dynamic) web pages.

Why the Term Hypertext & Markup Language?

The term 'Hypertext Markup Language' is composed of two main words: 'hypertext' and 'markup language.' 'Hypertext' refers to the linking of text with other documents, while 'markup language' denotes a language that utilizes a specific set of tags.

Thus, HTML is the practice of displaying text, graphics, audio, video, etc., in a certain way using special tags.

Note: Tags are meaningful texts enclosed in angle braces, like '<...>'. For example, the " tag. Each tag has a unique meaning and significance in building an HTML page, and it can influence the web page in various ways.

HTML Working

You must have heard of frontend and backend. Frontend refers to the visible part of a website or app that users interact with, like the tables, images, and buttons. It's built using languages like HTML, CSS, and JavaScript. The backend, on the other hand, handles behind-the-scenes operations like storing and processing data when users interact with the frontend. It uses languages like Python, Ruby, and Java. In essence, frontend is what users see, while backend manages the functionality.

HTML Page Structure

An HTML document is structured using a set of nested tags. Each tag is enclosed within `<...>` angle brackets and acts as a container for content or other HTML tags. Let's take a look at a basic HTML document str

Note: These are the essential elements for a basic HTML document: `<!DOCTYPE html>`, `<html>`, `<head>`, `<title>`, `</head>`, `<body>`, `</body>`, `</html>`

HTML Tags

If you want to build a beautiful website, tags are essential elements that help you achieve that.

An HTML tag acts as a container for content or other HTML tags. Tags are words enclosed within `<` and `>` angle brackets.

They serve as keywords that instruct the web browser on how to format and display the content.

There are some other tags like

Hyperlink and Media Tags

1. `<a>`: Anchor (used for links).
2. `<img>`: Image.
3. `<audio>`: Audio content.
4. `<video>`: Video content.

Form Tags

1. `<form>`: Form.

2. `<input>`: Input field.
3. `<textarea>`: Text area.
4. `<button>`: Button.
5. `<select>`: Dropdown list.
6. `<option>`: Options within a `<select>` or `<datalist>`.

Table Tags

1. `<table>`: Table.
2. `<tr>`: Table row.
3. `<td>`: Table data cell.
4. `<th>`: Table header cell.
5. `<thead>`: Table header group.
6. `<tbody>`: Table body group.
7. `<tfoot>`: Table footer group.

Semantic Tags

1. `<header>`: Header section.
2. `<footer>`: Footer section.
3. `<article>`: Article.
4. `<section>`: Section.

5. `<nav>`: Navigation.
6. `<aside>`: Sidebar content.

2.3 INTRODUCTION TO CSS

What is CSS?

CSS stands for Cascading Style Sheets. It is a stylesheet language that is used to describe the visual presentation of a web page written in HTML (Hypertext Markup Language). HTML creates the structure of the page, while CSS adds styling to that structure.

Why use CSS?

CSS is essential for creating visually appealing and responsive websites. Here are some key reasons to use CSS:

- **Enhances Visual Appeal:** Transforms plain HTML into vibrant, professional-looking web pages.
- **Improves User Experience:** Makes websites intuitive and easy to navigate.
- **Enables Responsive Design:** Adapts layouts for various devices, from mobile phones to desktops.
- **Supports Interactivity:** Adds dynamic effects like hover states, transitions, and animations.
- **Promotes Reusability:** Allows the same styles to be applied across multiple pages, saving time and effort.

Key Features of CSS:

- Styles and layouts of web pages.
- Works alongside HTML and XML documents.
- Enables responsive design for different screen sizes.

- Supports interactive effects like hover states and animations.
- CSS is now modularized, with ongoing updates rather than version numbers.
- Allows reusability of the same rules across multiple HTML documents.

How CSS Works?

CSS is a stylesheet language that styles HTML elements by applying visual properties like colors, fonts, margins, and layouts. It works by selecting specific elements in the DOM and defining their appearance using rules.

CSS Rule Structure

A CSS rule consists of three parts:

- Selector: Targets an HTML element, class, or ID (e.g., `h1`, `.class-name`, `#id-name`).
- Property: Defines the style attribute to modify (e.g., `color`, `font-size`).
- Value: Specifies the setting for the property (e.g., `blue`, `24px`).

HOW CSS WORKS ?

CSS transforms a webpage by interacting with the DOM through a series of steps.

- The user types the URL and clicks enter.
- The browser makes a fetch request to the server.
- HTML is fetched from the server.
- HTML is converted into a DOM. In the DOM, each tag is considered a **node**.
- The browser fetches all the related files and assets that are linked to that HTML, such as external CSS, fonts, images, etc.
- The browser then parses the CSS and groups it based on the selectors, which can be tags.

- Each CSS is attached to its respective node. In this phase, CSS gets attached to its respective node. This is called a **render tree**.
- The render tree is the well-structured, well-arranged **DOM** node that will appear on the screen.
- The well-structured, custom-designed website is presented on the screen. This is called **painting**.

Ways to Add CSS

There are three different ways to add CSS to an HTML page, which are:

1. Inline CSS
2. Internal CSS
3. External CSS

Inline CSS

Inline CSS is used to add custom properties to specific elements. The added style will only reflect on that particular element only.

Implementation To use inline CSS, insert the style attribute within the HTML element's opening tag.

Code snippet:

```
<h1 style="color: purple;">HEADING1</h1>
```

```
<h2>HEADING2</h2>
```

Internal CSS

Internal CSS is used to apply custom style to multiple elements on the same page. The style can be used throughout the HTML page.

Implementation Internal CSS is defined in a **style block**, which will be inside the **head section**.

Code snippet:

```
<head>

  <style>

    p {

      color: red;

    }

  </style>

</head>

<body>

  <h1>HEADING!</h1>

  <p>Paragraph</p>

</body>
```

External CSS

External CSS works similarly to internal CSS but with a twist. Instead of adding the styles within the HTML file, we create a separate file with a **.css** extension. This file will hold all the styling details. Then, we link this file to the HTML page, giving it the instructions on how to look.

There is a new `<link>` tag in the head section, and this link tag has `rel` and `href` properties.

The following points will explain each keyword's meaning:

- 1) **<link>**: This tag is used to create links between different resources, like stylesheets, fonts, and more. In our case, we are using a link tag to link the CSS file with the HTML file.
- 2) **rel="stylesheet"**: rel stands for relationship, this defines the type of relationship between the HTML document and the linked resource. When set to "stylesheet", it specifies that the linked resource is a stylesheet that will be used to style the HTML content.
- 3) **href="style.css"**: The href attribute stands for "hypertext reference." It specifies the path or URL to the external resource we want to link. In this case, it's the path to the external CSS file called "style.css".

Code snippet:

```
<html lang="en">  
  
<head>  
  
  <title>Opinder</title>  
  
  <link rel="stylesheet" href="style.css">  
  
</head>  
  
<body>  
  
  <p>Paragraph</p>  
  
</body>  
  
</html>
```

External CSS File:

```
p {  
  
  color: red;  
  
}
```

2.4 INTRODUCTION TO JS

JavaScript is what makes a website interactive. It's like adding electricity to the house. It lets you click buttons to open doors, turn on lights, or even play music. So, without JavaScript, a website would be like a house where nothing really happens—you could look at it.

For Example :- Imagine a website as a house. HTML is like the bricks and walls that give it structure. CSS is like the paint and decorations that make it look nice. But without JavaScript, the house won't have any lights or running water—nothing would work or move.

What is JavaScript?

- It is a scripting language for web pages.
- It is used to add interactivity and dynamic effects to web pages.
- ".js" is the extension.
- Nowadays used in server-side development.
- **Frontend Frameworks:** FRONTEND, Angular, Vue.
- **Backend Frameworks:** Express, Node.

How to Execute JavaScript?

JavaScript can be executed right inside one's browser. You can open the console and start writing there.

Yet another way to execute JavaScript is by inserting it inside the `<script>` tag of an HTML document.

What are Variables?

In JavaScript, variables are used to store data. They are an essential part of any programming language, as they allow you to store, retrieve, and manipulate data in your programs.

There are a few different ways to declare variables in JavaScript, each with its own syntax and rules. In this blog post, we'll take a look at the different types of variables available in JavaScript and how to use them.

Declaring Variables

To declare a variable in JavaScript, you use the `var` keyword followed by the name of the variable.

For example: `var x;`

This declares a variable called `x` but does not assign any value to it. You can also assign a value to a variable when you declare it, like this: `var x = 10;`

In JavaScript, you can also use the `let` and `const` keywords to declare variables. The `let` keyword is used to declare a variable that can be reassigned later, while the `const` keyword is used to declare a variable that cannot be reassigned. For example:

```
let y = 20;
```

```
const z = 30;
```

Data Types

In JavaScript, there are several data types that you can use to store different types of data. Some common data types include:

- Numbers (e.g. 10, 3.14)
- Strings (e.g. "hello", 'world')

- Booleans (e.g. true, false)
- Arrays (e.g. [1, 2, 3])
- Objects (e.g. { name: "John", age: 30 })

Operators and Expressions

Operators in JavaScript are symbols that perform specific operations on one or more operands (values or variables). For example, the addition operator (+) adds two operands together and the assignment operator (=) assigns a value to a variable.

There are several types of operators in JavaScript, including:

- Arithmetic operators (+, -, *, /, %)
- Comparison operators (>, <, >=, <=, ==, !=)
- Logical operators (&&, ||, !)
- Assignment operators (=, +=, -=, *=, /=)
- Conditional (ternary) operator (?:)

var vs let vs const

In JavaScript, there are three ways to declare variables: var, let, and const. Each of these keywords has its own rules and uses, and it is important to understand the differences between them in order to write effective and maintainable code.

CHAPTER 3

RESULTS AND DISCUSSION/OBSERVATION

This chapter presents the final outcome of the Portfolio project, detailing the functionalities achieved through the implementation of FRONTEND. It also includes a critical discussion of the project's observations, limitations, and potential for future development.

3.1 Results

This section objectively presents the final, working application and its features as realized in the codebase.

1. Successful Implementation of a Component-Based Architecture

The Portfolio application was successfully developed as a single-page application (SPA) using HTML ,Javascript,CSS. The project's source code is logically structured into four primary, reusable components, which are assembled in the main Portfolio file:

- **Header.Portfolio:** A presentational component that renders the static header, including the application's logo and title, ensuring consistent branding across the user experience.

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4   <head>
5     <meta charset="UTF-8" />
6     <meta http-equiv="X-UA-Compatible" content="IE=edge" />
7     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
8     <meta name="google-adense-account" content="ca-pub-7199918361852779" />
9
10    <!-- Favicon -->
11    <link rel="shortcut icon" href="assets/img/favicon.png" type="image/x-icon" />
12
13    <!-- Realism CSS -->
14    <link href="https://cdn.jsdelivr.net/npm/realismcss@3.0/fonts/realism.css" rel="stylesheet" />
15
16    <!-- Main CSS -->
17    <link rel="stylesheet" href="assets/css/main.css" />
18
19    <!-- Google AdSense -->
20    <script async src="https://pagead2.googlesyndication.com/pagead/js/adsbygoogle.js"></script>
21    <script>
22      (adsbygoogle = window.adsbygoogle || []).push({
23        google_ad_client: "ca-pub-7199918361852779",
24        enable_page_level_ads: true
25      });
26    </script>
27  </head>
28
29  <body>
30    <!-- Header -->
31    <div class="header" id="header">
32      <div class="nav-container">
33        <a href="#" class="nav-link" />
34        <a href="#" class="nav-link" />
35        <a href="#" class="nav-link" />
36        <a href="#" class="nav-link" />
37        <a href="#" class="nav-link" />
38      </div>
39    </div>
40  </body>
41</html>
```

Figure 3.1 Code of Header (Portfolio)

Figure 3.2 Header Output (Portfolio)

- **Intro:** A static component that provides a welcoming message and a clear, concise explanation of the application's purpose to the user.

```

<!-- ===== HOME ===== -->
<section class="home section" id="home">
  <div class="home__container container grid section__border">
    <div class="home__data grid">
      <h1 class="home__title">
        Hi, I'm Sidak <br />
        Frontend Developer <br />
        based on India <br />
      </h1>
      <div class="home__blob grid">
        <div class="home__perfill">
          
        </div>
        
        
      </div>
      <ul class="home__social">
        <a href="#" target="_blank" class="home__social-link">
          <i class="ri-linkedin-box-fill"></i>
        </a>
        <a href="#" target="_blank" class="home__social-link">
          <i class="ri-github-fill"></i>
        </a>
        <a href="#" target="_blank" class="home__social-link">
          <i class="ri-dribbble-line"></i>
        </a>
      </ul>
    </div>
  </div>
</section>

```

Figure 3.3 Code of Intro (Portfolio)

Hi, I'm Sidak
Frontend Developer
based on India

Figure 3.4 Intro Output (Portfolio)

- **Main:** This is the core interactive component of the application.



Figure 3.5 Main Output (Portfolio)

```
<!--===== FOOTER =====>
<footer class="footer">
  <div class="footer__container container">
    <h1 class="footer__title">Sidak</h1>
    <p>Frontend Developer</p>
    <ul class="footer__list">
      <li><a href="#home" class="footer__link"> Home</a></li>
      <li><a href="#skills" class="footer__link">Skills</a></li>
      <li><a href="#projects" class="footer__link">Projects</a></li>
    </ul>
```

Figure 3.6 Code of Footer (Portfolio)

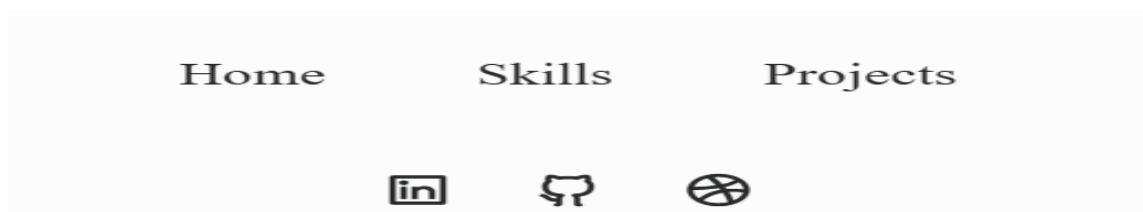


Figure 3.6 Footer Output (Portfolio)

OVERALL OVERVIEW OF THE PROJECT

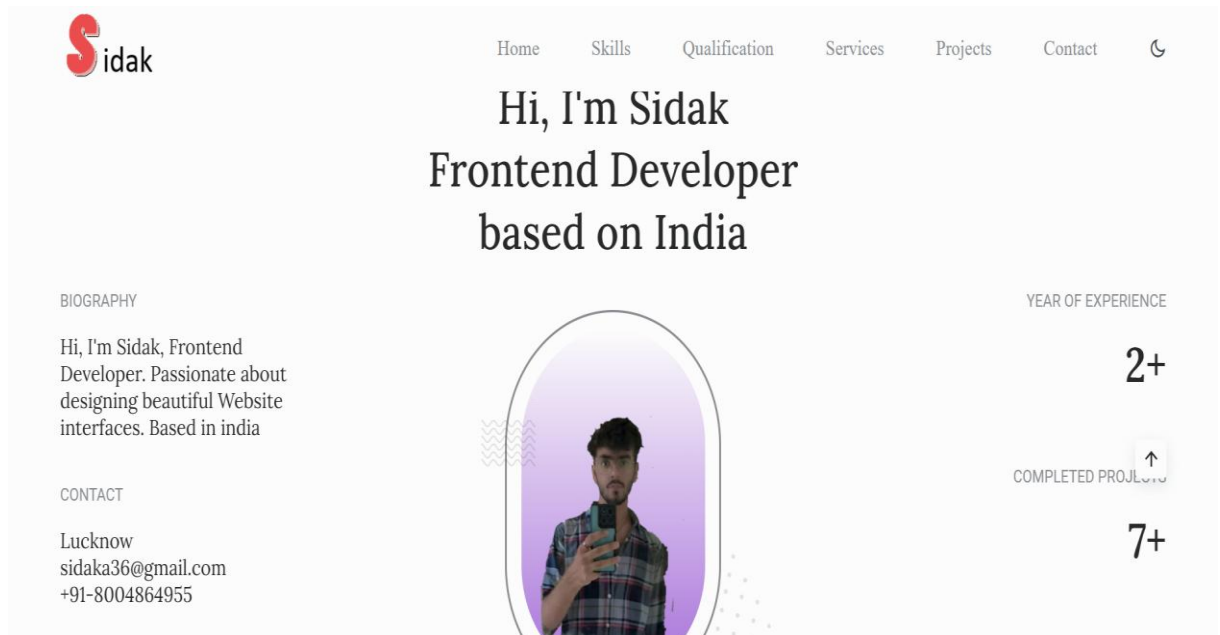


Figure 3.7 OVERVIEW OF PROJECT (Portfolio)

CHAPTER 4

CONCLUSION

The **Web development** training and the subsequent development of the **Portfolio** project provided a comprehensive, hands-on learning experience, successfully bridging the gap between theoretical concepts and practical application. This journey culminated in the creation of a functional, single-page web application, demonstrating a solid understanding of modern frontend development principles.

Synthesis of Learning and Application

The training laid a strong foundation in the core principles of Web development, which were directly applied and solidified during the project's development.

- **Component-Based Architecture:** The training's emphasis on breaking down UIs into small, reusable pieces was the guiding principle for structuring Portfolio. The application was logically divided into components like Header, Intro, Main, and Footer, resulting in clean, organized, and maintainable code.
- **State Management:** The most critical concept, state management, was mastered through the use of the `useState` hook. This was fundamental to the project's interactivity, allowing the application to dynamically handle the user's list of ingredients and display a recipe without reloading the page.

Key Takeaways and Skills Gained

This experience has equipped me with a set of in-demand, practical skills that are essential for a career in frontend development:

- **Modern Frontend Development:** Proficiency in building dynamic, single-page applications using the Web development.
- **Effective State Management:** The ability to manage and manipulate component state to create interactive and responsive user interfaces.
- **UI/UX Implementation:** The skill to translate project requirements and wireframes into a functional, aesthetically pleasing, and user-friendly interface using X and CSS.
- **Problem-Solving and Debugging:** The competence to identify, troubleshoot, and solve common development challenges within a FRONTEND environment, such as managing data flow between components.

Project Outcome and Future Outlook

The **PORTFOLIO** project successfully met its primary objective of creating a working Portfolio. While it effectively demonstrates the core functionalities of a HTML,CSS,Javascript ,application, it also serves as a foundation for future enhancements. Acknowledging its current limitation—the Introduction—provides a clear roadmap for improvement, such as implementing a true ingredient-matching algorithm and integrating with a third-party API.

Ultimately, this training and project have been invaluable. They have not only solidified my technical skills in Web development but have also instilled the confidence to design, develop, and deploy complex frontend applications, marking a significant step forward in my journey as a web developer.

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